problems 3, 5, and 6

```
clear; clc; close all;

repoRoot = localResolveRepoRoot();
origDir = pwd;
cleanupDir = onCleanup(@() cd(origDir)); %#ok<NASGU>
cd(repoRoot);
addpath(genpath(fullfile(repoRoot, 'scripts')));
run(fullfile(repoRoot, 'scripts', 'control',
'setup_eul_controller_demo.m'));
```

```
Trim point loaded into workspace:
    front_collective_rpm    rear_collective_rpm    delta_e_deg    theta_deg    vinf_mps
    ____________________    ___________________    ___________    _________    ________

           1181.4                     0                -15.06        2.6325         70

Controller gains:
        V: [1×1 struct]
      phi: [1×1 struct]
    theta: [1×1 struct]

Sensor settings:
    airspeed: [1×1 struct]
       alpha: [1×1 struct]
        beta: [1×1 struct]
       omega: [1×1 struct]
       accel: [1×1 struct]
         pos: [1×1 struct]
         vel: [1×1 struct]
         eul: [1×1 struct]

Truth-memory initial conditions:
                 pos_NED: [3×1 double]
               V_B_truth: [3×1 double]
               V_E_truth: [3×1 double]
               eul_truth: [3×1 double]
             omega_truth: [3×1 double]
              C_NB_truth: [3×3 double]
              V_BA_truth: [3×1 double]
              vinf_truth: 70
             alpha_truth: 0.045946
              beta_truth: 0
    specific_force_truth: [3×1 double]

Logged bus indices:
       sensor_bus: [1×1 struct]
    plant_cmd_bus: [1×1 struct]

Bus objects:
    base_set_point: 'BaseSetPointBus'
       cmd_profile: 'CmdProfileBus'
     command_delta: 'CommandDeltaBus'
         truth_bus: 'TruthBus'
```

```
controller_sample_time = 0.01;
modelName = 'Brown_6DOF_Sim_Wrapper';
```

gains

```
gains.V.Kp = 300;
gains.V.Ki = 50;
gains.theta.Kp = 2.10;
gains.theta.Kd = 0.10;
gains.phi.Kp = 2;
gains.phi.Kd = -2;

% Step commands
step.longitudinal.time = 3.0;
step.longitudinal.d_vinf_cmd = 5.0;
step.lateral.time = 3.0;
step.lateral.d_phi_cmd = deg2rad(5.0);

gainSummary = table( ...
    gains.V.Kp, gains.V.Ki, ...
    gains.theta.Kp, gains.theta.Kd, ...
    gains.phi.Kp, gains.phi.Kd, ...
    'VariableNames', { ...
    'V_Kp', 'V_Ki', ...
    'theta_Kp', 'theta_Kd', ...
    'phi_Kp', 'phi_Kd'})
```

gainSummary = 1×6 table

	V_Kp	V_Ki	theta_Kp	theta_Kd	phi_Kp	phi_Kd
1	300	50	2.1	0.1	2	-2

```
trimSummary = table( ...
    base_set_point.vinf_cmd, ...
    rad2deg(base_set_point.euler_cmd(2)), ...
    base_set_point.front_collective_rpm, ...
    rad2deg(base_set_point.delta_e_cmd), ...
    'VariableNames', { ...
    'trim_vinf_cmd_mps', ...
    'trim_theta_cmd_deg', ...
    'trim_front_collective_rpm', ...
    'trim_delta_e_deg'})
```

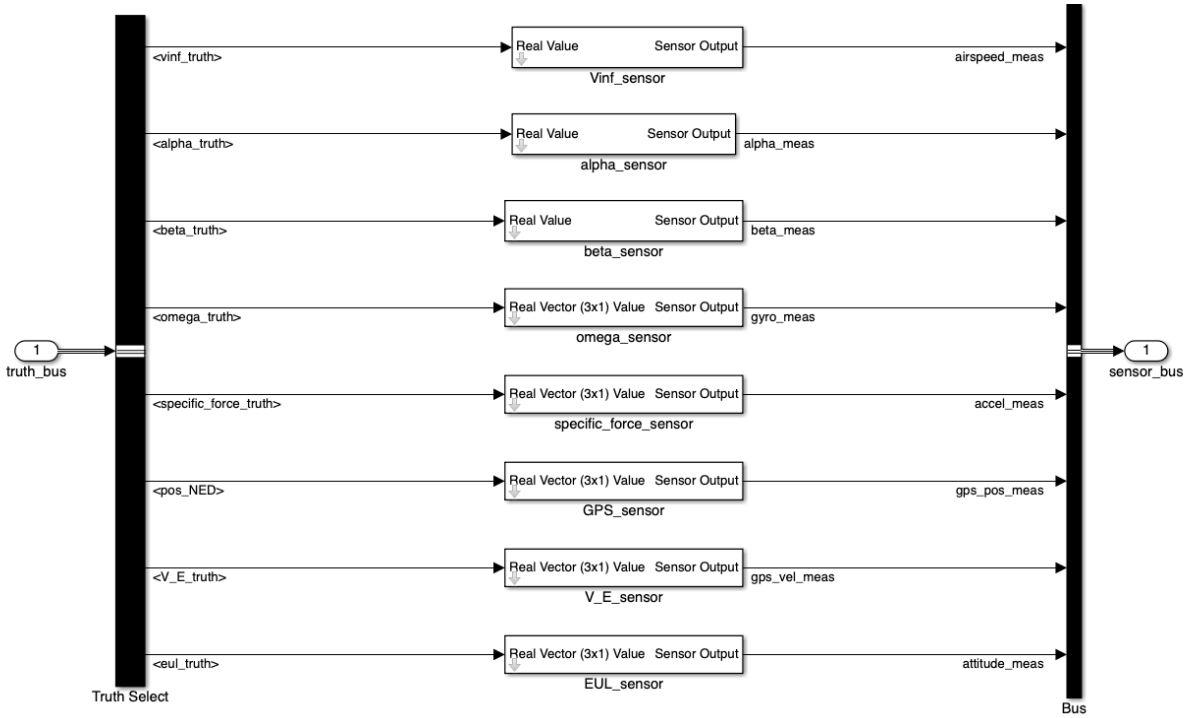
trimSummary = 1×4 table

...

	trim_vinf_cmd_mps	trim_theta_cmd_deg	trim_front_collective_rpm
1	70	2.6325	1181.4

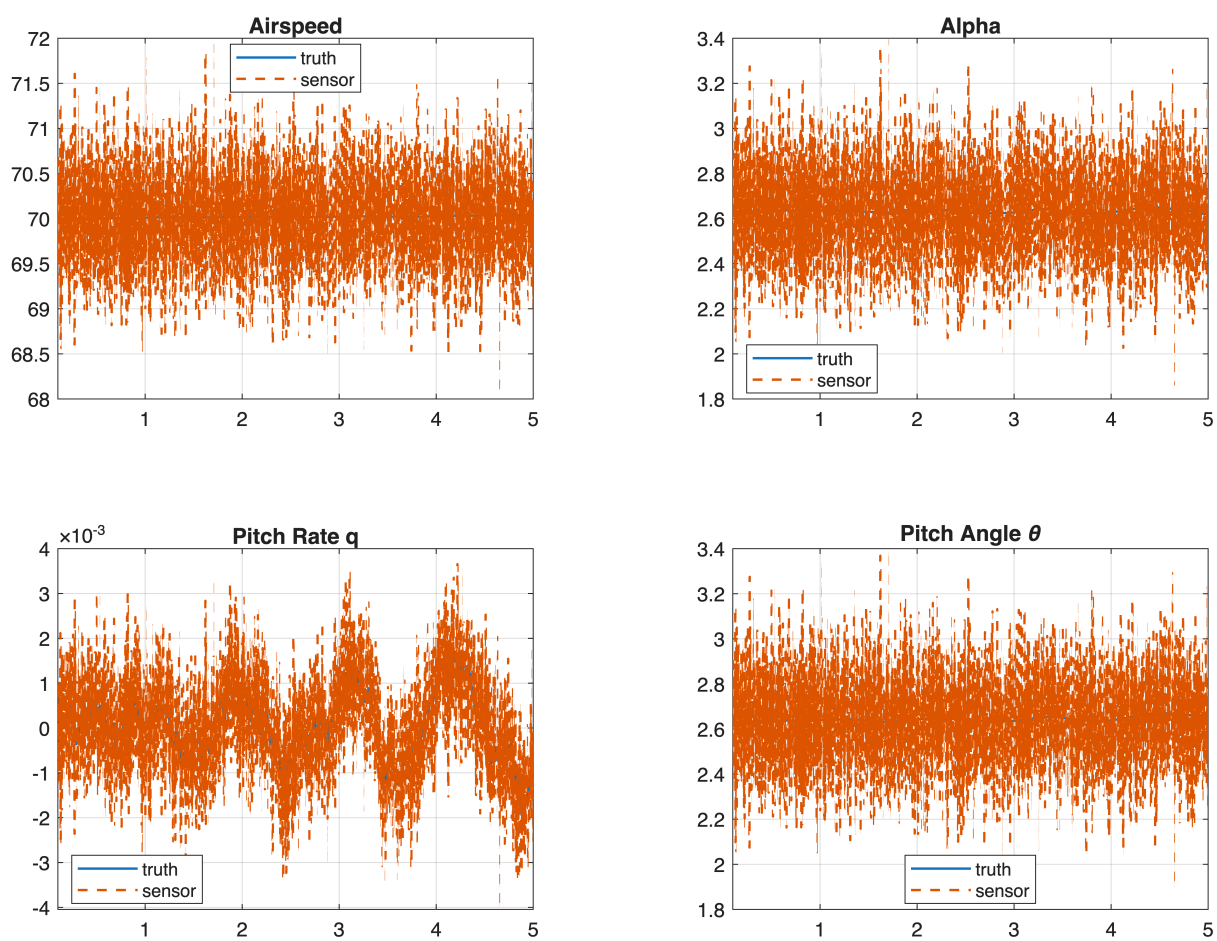
problem3

sensors are built into the simulink model




```
cmd_profile.vinf.enable = false;
cmd_profile.phi.enable = false;
cmd_profile.theta.enable = false;

simOut3 = localRunSim(modelName, 5);

[tVinfTruth3, vinfTruth3] = localSignal(simOut3.get('vinf_truth'));
[tAlphaTruth3, alphaTruth3] = localSignal(simOut3.get('alpha_truth'));
[tEulTruth3, eulTruth3] = localSignal(simOut3.get('eul_truth'));
[tOmegaTruth3, omegaTruth3] = localSignal(simOut3.get('omega_truth'));

sensor3 = localSensorBus(simOut3.get('sensor_bus'), log_idx.sensor_bus);

figure('Color', 'w', 'Position', [100 100 1000 720]);
tiledlayout(2,2);

nexttile;
plot(tVinfTruth3, vinfTruth3, 'LineWidth', 1.2); hold on;
plot(sensor3.t, sensor3.airspeed, '--', 'LineWidth', 1.2);
grid on;
xlim([0.1 5]);
title('Airspeed');
legend('truth', 'sensor', 'Location', 'best');

nexttile;
plot(tAlphaTruth3, rad2deg(alphaTruth3), 'LineWidth', 1.2); hold on;
plot(sensor3.t, rad2deg(sensor3.alpha), '--', 'LineWidth', 1.2);
grid on;
xlim([0.1 5]);
```

```
title('Alpha');
legend('truth', 'sensor', 'Location', 'best');

nexttile;
plot(tOmegaTruth3, omegaTruth3(:,2), 'LineWidth', 1.2); hold on;
plot(sensor3.t, sensor3.gyro(:,2), '--', 'LineWidth', 1.2);
grid on;
xlim([0.1 5]);
title('Pitch Rate q');
legend('truth', 'sensor', 'Location', 'best');

nexttile;
plot(tEulTruth3, rad2deg(eulTruth3(:,2)), 'LineWidth', 1.2); hold on;
plot(sensor3.t, rad2deg(sensor3.attitude(:,2)), '--', 'LineWidth', 1.2);
grid on;
xlim([0.1 5]);
title('Pitch Angle \theta');
legend('truth', 'sensor', 'Location', 'best');
```



```
sensorSummary = table( ...
    vinfTruth3(end), sensor3.airspeed(end), ...
    rad2deg(alphaTruth3(end)), rad2deg(sensor3.alpha(end)), ...
    omegaTruth3(end,2), sensor3.gyro(end,2), ...
    rad2deg(eulTruth3(end,2)), rad2deg(sensor3.attitude(end,2)), ...
    'VariableNames', { ...
    'vinf_truth_end', 'vinf_sensor_end', ...
    'alpha_truth_end_deg', 'alpha_sensor_end_deg', ...
    'q_truth_end', 'q_sensor_end', ...
    'theta_truth_end_deg', 'theta_sensor_end_deg'})
```

sensorSummary = 1×8 table

...

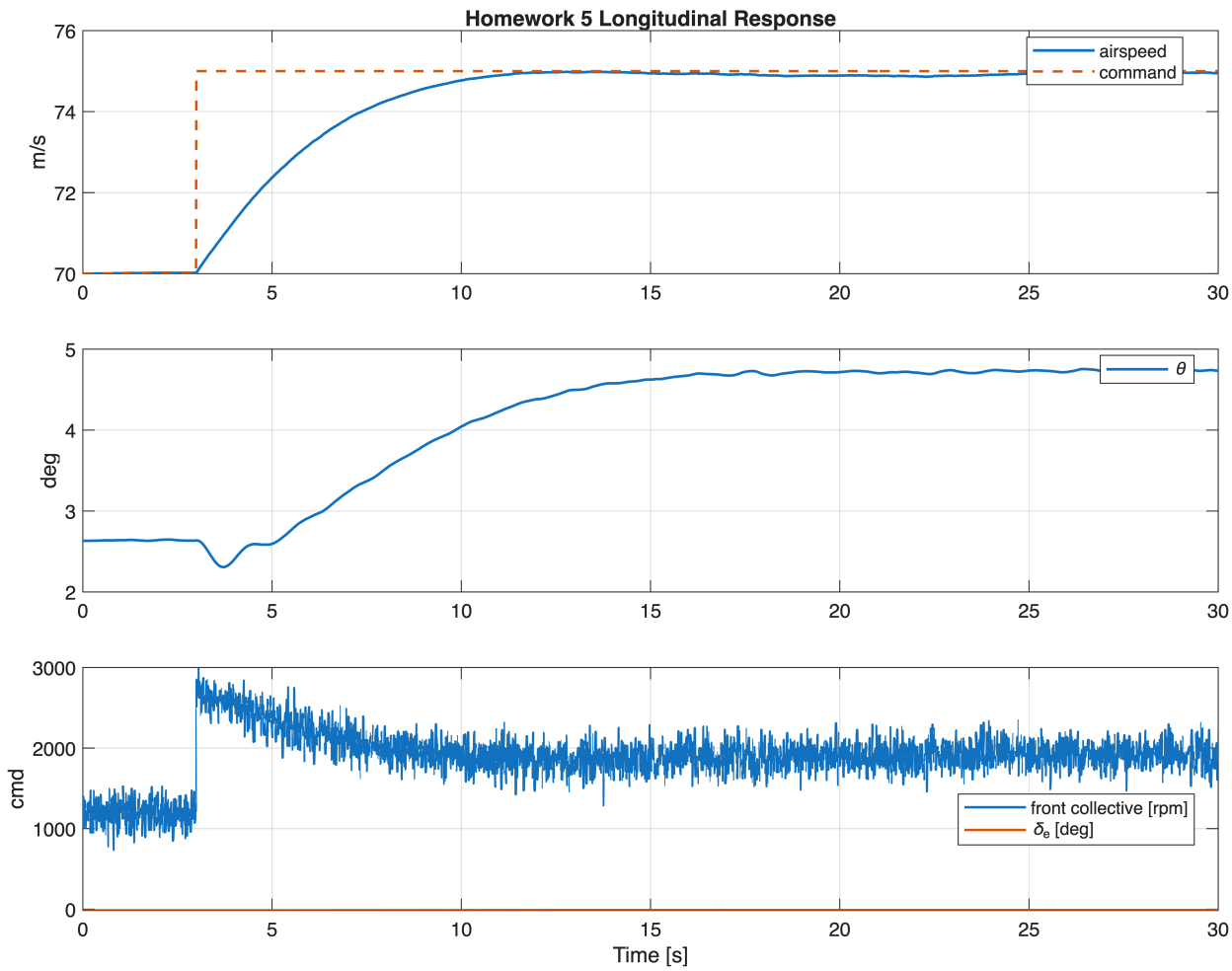
	vinf_truth_end	vinf_sensor_end	alpha_truth_end_deg	alpha_sensor_end_deg
1	70.038	70.392	2.6197	2.7612

problem 5

I am using airspeed feedback here. The command steps up by 5 m/s at 3 s.

```
cmd_profile.vinf.enable = true;
cmd_profile.vinf.time = step.longitudinal.time;
cmd_profile.vinf.magnitude = step.longitudinal.d_vinf_cmd;
cmd_profile.phi.enable = false;
cmd_profile.theta.enable = false;

simOut5 = localRunSim(modelName, 30);

[tVinf5, vinfTruth5] = localSignal(simOut5.get('vinf_truth'));
[tCmd5, vinfCmd5] = localSignal(simOut5.get('vinf_cmd'));
[tEul5, eulTruth5] = localSignal(simOut5.get('eul_truth'));
plantCmd5 = localPlantCmdBus(simOut5.get('plant_cmd_bus'),
log_idx.plant_cmd_bus);

beforeIdx5 = localBeforeStepIndex(tVinf5, cmd_profile.vinf.time);

figure('Color', 'w', 'Position', [100 100 950 700]);
tiledlayout(3,1);

nexttile;
plot(tVinf5, vinfTruth5, 'LineWidth', 1.3); hold on;
plot(tCmd5, vinfCmd5, '--', 'LineWidth', 1.2);
grid on;
ylabel('m/s');
title('Homework 5 Longitudinal Response');
legend('airspeed', 'command', 'Location', 'best');

nexttile;
plot(tEul5, rad2deg(eulTruth5(:,2)), 'LineWidth', 1.3);
grid on;
```

```
ylabel('deg');
legend('\theta', 'Location', 'best');

nexttile;
plot(plantCmd5.t, plantCmd5.front_collective_rpm, 'LineWidth', 1.2); hold
on;
plot(plantCmd5.t, rad2deg(plantCmd5.delta_e_cmd), 'LineWidth', 1.2);
grid on;
xlabel('Time [s]');
ylabel('cmd');
legend('front collective [rpm]', '\delta_e [deg]', 'Location', 'best');
```




```
longSummary = table( ...
    vinfTruth5(beforeIdx5), vinfCmd5(beforeIdx5), ...
    vinfTruth5(end), vinfCmd5(end), ...
    plantCmd5.front_collective_rpm(end),
rad2deg(plantCmd5.delta_e_cmd(end)), ...
    'VariableNames', { ...
    'vinf_before_step', 'vinf_cmd_before_step', ...
```

```
    'vinf_end', 'vinf_cmd_end', ...
    'front_collective_end_rpm', 'delta_e_end_deg'})
```

longSummary = 1×6 table

...

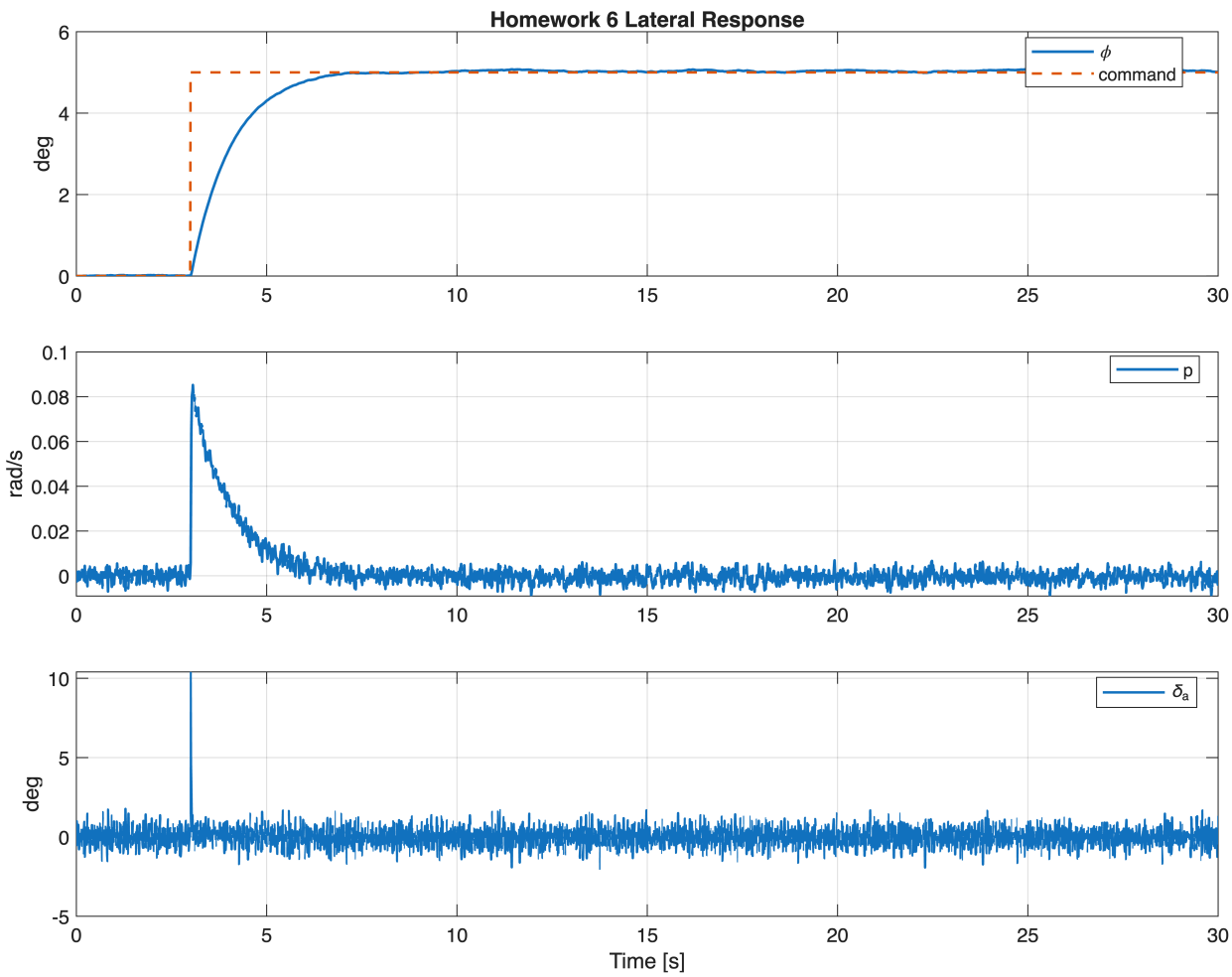
	vinf_before_step	vinf_cmd_before_step	vinf_end	vinf_cmd_end
1	70.021	70	74.95	75

problem 6 - lateral set-point hold and step

I am using bank-angle feedback here. The command steps up by 5 deg at 3 s.

```
cmd_profile.vinf.enable = false;
cmd_profile.phi.enable = true;
cmd_profile.phi.time = step.lateral.time;
cmd_profile.phi.magnitude = step.lateral.d_phi_cmd;
cmd_profile.theta.enable = false;

simOut6 = localRunSim(modelName, 30);

[tEul6, eulTruth6] = localSignal(simOut6.get('eul_truth'));
[tCmd6, eulerCmd6] = localSignal(simOut6.get('euler_cmd'));
[tOmega6, omegaTruth6] = localSignal(simOut6.get('omega_truth'));
plantCmd6 = localPlantCmdBus(simOut6.get('plant_cmd_bus'),
log_idx.plant_cmd_bus);

beforeIdx6 = localBeforeStepIndex(tEul6, cmd_profile.phi.time);

figure('Color', 'w', 'Position', [100 100 950 700]);
tiledlayout(3,1);

nexttile;
plot(tEul6, rad2deg(eulTruth6(:,1)), 'LineWidth', 1.3); hold on;
plot(tCmd6, rad2deg(eulerCmd6(:,1)), '--', 'LineWidth', 1.2);
grid on;
ylabel('deg');
title('Homework 6 Lateral Response');
legend('\phi', 'command', 'Location', 'best');

nexttile;
plot(tOmega6, omegaTruth6(:,1), 'LineWidth', 1.3);
grid on;
ylabel('rad/s');
legend('p', 'Location', 'best');

nexttile;
plot(plantCmd6.t, rad2deg(plantCmd6.delta_a_cmd), 'LineWidth', 1.2);
grid on;
xlabel('Time [s]');
ylabel('deg');
```

```
legend('\delta_a', 'Location', 'best');
```




```
latSummary = table( ...
    rad2deg(eulTruth6(beforeIdx6,1)), rad2deg(eulerCmd6(beforeIdx6,1)), ...
    rad2deg(eulTruth6(end,1)), rad2deg(eulerCmd6(end,1)), ...
    rad2deg(plantCmd6.delta_a_cmd(end)), ...
    'VariableNames', { ...
    'phi_before_step_deg', 'phi_cmd_before_step_deg', ...
    'phi_end_deg', 'phi_cmd_end_deg', ...
    'delta_a_end_deg'})
```

latSummary = 1×5 table

...

	phi_before_step_deg	phi_cmd_before_step_deg	phi_end_deg	phi_cmd_end_deg
1	0.0098834	0	5.0134	5

```
function [t, y] = localSignal(sig)
t = sig.time(:);
vals = sig.signals.values;
y = squeeze(vals);

if isvector(y)
    y = y(:);
    return;
end

n = numel(t);
if size(y,1) == n
    return;
end

if size(y, ndims(y)) == n
    y = reshape(y, [], n).';
else
    y = reshape(y, n, []);
end
end

function sensor = localSensorBus(sig, idx)
[t, raw] = localSignal(sig);

sensor = struct();
sensor.t = t;
sensor.airspeed = raw(:, idx.airspeed);
sensor.alpha = raw(:, idx.alpha);
sensor.beta = raw(:, idx.beta);
sensor.gyro = raw(:, idx.gyro);
sensor.accel = raw(:, idx.accel);
sensor.gps_pos = raw(:, idx.gps_pos);
sensor.gps_vel = raw(:, idx.gps_vel);
sensor.attitude = raw(:, idx.attitude);
end

function cmd = localPlantCmdBus(sig, idx)
[t, raw] = localSignal(sig);

cmd = struct();
cmd.t = t;
cmd.motor_rpm_cmd = raw(:, idx.motor_rpm_cmd);
cmd.tilt_angles_cmd = raw(:, idx.tilt_angles_cmd);
cmd.front_collective_rpm = raw(:, idx.front_collective_rpm);
cmd.rear_collective_rpm = raw(:, idx.rear_collective_rpm);
cmd.delta_f_cmd = raw(:, idx.delta_f_cmd);
cmd.delta_a_cmd = raw(:, idx.delta_a_cmd);
cmd.delta_e_cmd = raw(:, idx.delta_e_cmd);
cmd.delta_r_cmd = raw(:, idx.delta_r_cmd);
```

```
end

function idx = localBeforeStepIndex(t, stepTime)
idx = find(t < stepTime, 1, 'last');
if isempty(idx)
    idx = 1;
end
end

function simOut = localRunSim(modelName, stopTime)
warnState = warning;
cleanupWarn = onCleanup(@() warning(warnState));
warning('off', 'all');
simOut = sim(modelName, 'StopTime', num2str(stopTime),
'ReturnWorkspaceOutputs', 'on');
end

function repoRoot = localResolveRepoRoot()
candidates = {};

try
    thisPath = mfilename('fullpath');
    if ~isempty(thisPath)
        candidates{end+1} = fileparts(thisPath);
    end
catch
end

try
    activeFile = matlab.desktop.editor.getActiveFilename;
    if ~isempty(activeFile)
        candidates{end+1} = fileparts(activeFile);
    end
catch
end

candidates{end+1} = pwd;

for i = 1:numel(candidates)
    root = candidates{i};
    if exist(fullfile(root, 'scripts', 'control',
'setup_eul_controller_demo.m'), 'file')
        repoRoot = root;
        return;
    end
end

error('Could not find the repo root that contains scripts/control/
setup_eul_controller_demo.m.');
end
```